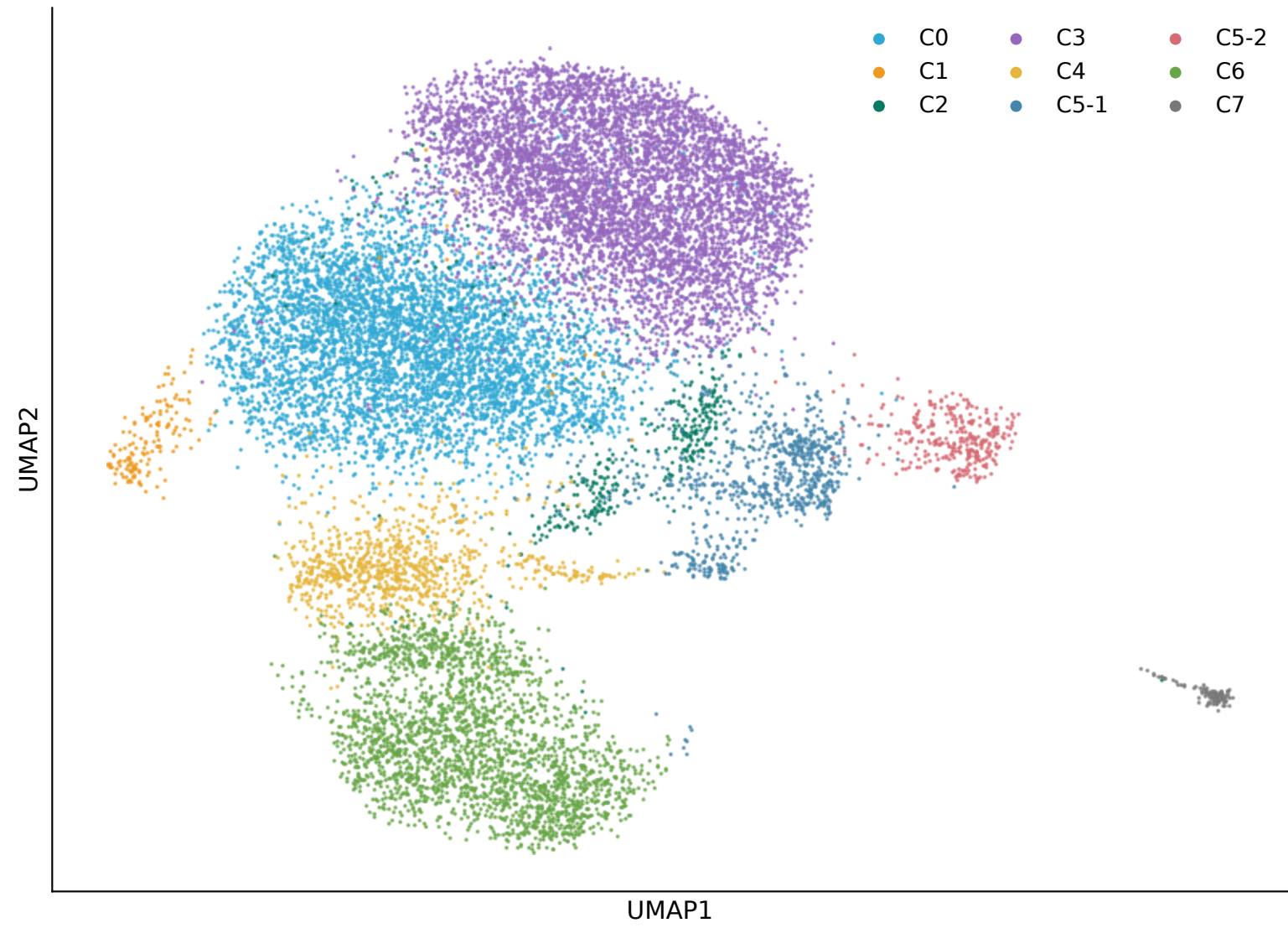


iNKT cytotoxicity and Gene Ontology

Paper-guided analysis on the latest local data | 19 September 2026



15,532 cells | 10,670 measured genes

PAGER: gene modules and temporal profiles

3 tissues | Ctrl and T2 | 9 refined clusters

GAFA: cluster-specific GO enrichment

Primary contrasts: tissue x original cluster

Sensitivity: stable tissue reclustering + QC matching

Analysis design and reference mapping

Methods adapted to mouse iNKT; biological conclusions are derived from these data.

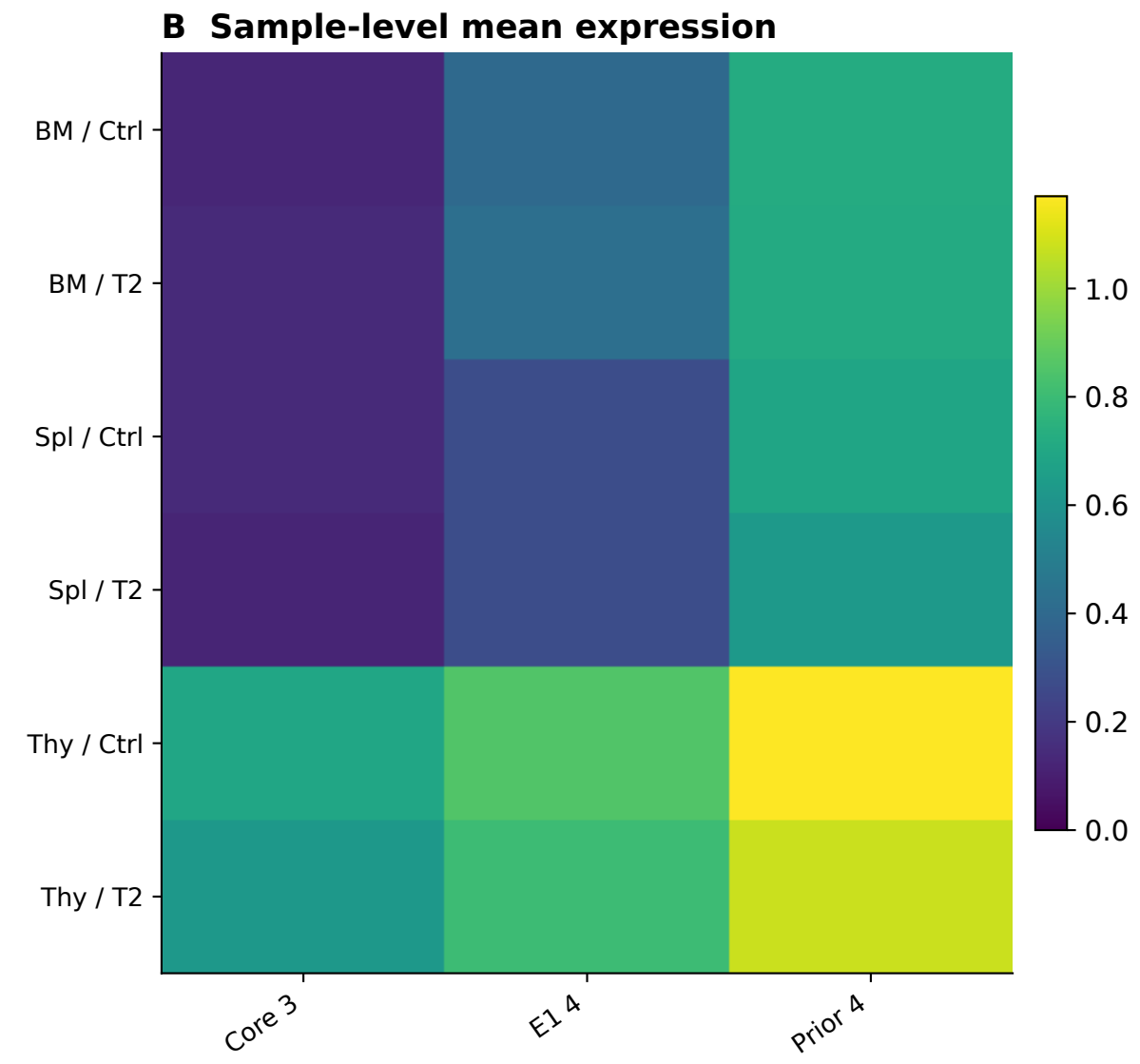
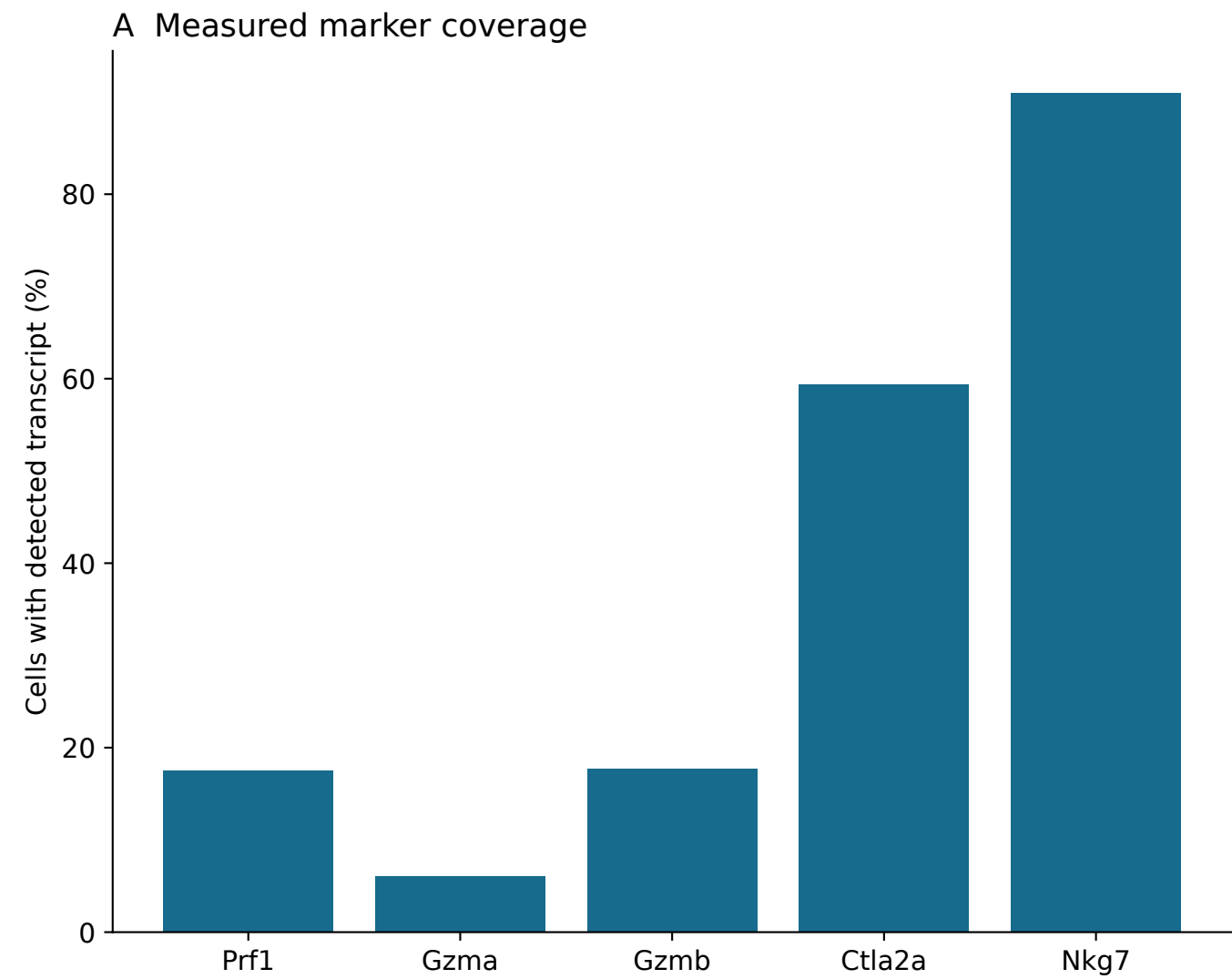
Analysis	Reference	Input / definition	Implementation
Cytotoxicity	PAGER Fig. 5C/D	Core: Prf1, Gzma, Gzmb; E1: +Ctla2a	Mean log1p expression; matched-control score sensitivity
GO cluster identity	GAFA Fig. 4c	Cluster vs rest within each tissue	Wilcoxon, gene BH <= .05 and log2FC >= .25
GO treatment response	GAFA functional comparison	T2-up and Ctrl-up genes tested separately	Hypergeometric; full eligible GO BP/MF/CC families
Local trajectories	PAGER Fig. 5D	Recompute diffusion map on tissue-specific graphs	Precursor-rooted DPT; 20 shared bins; 3-root sensitivity
Robustness	Adaptation to this dataset	Stable clusters; five QC-matched resamples	Effects and direction consistency; no animal replication

GO uses official mouse annotations. An enrichment of upregulated genes does not prove pathway activation.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Cytotoxic gene coverage and scoring

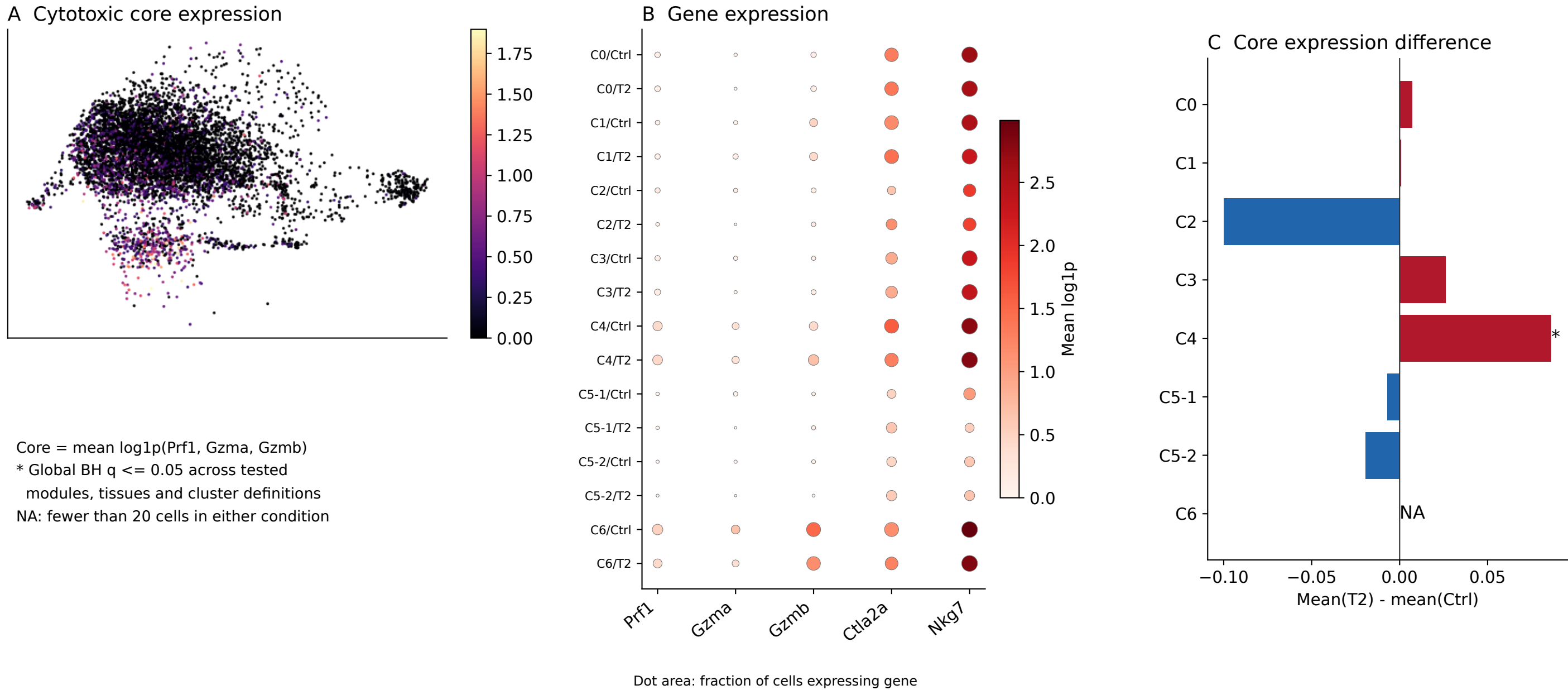
Primary endpoint: mean log-normalized expression of Prf1 / Gzma / Gzmb.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: cytotoxic expression and condition effects

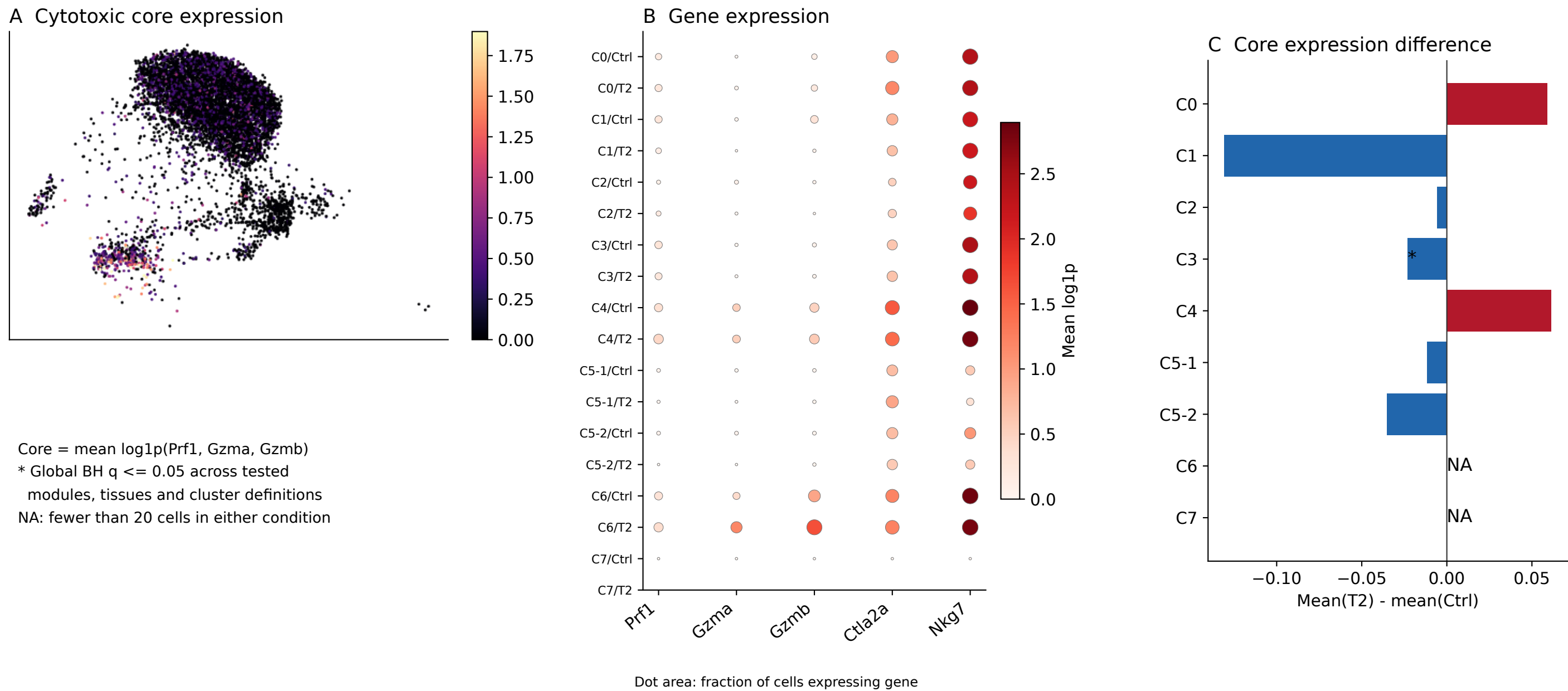
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Spleen: cytotoxic expression and condition effects

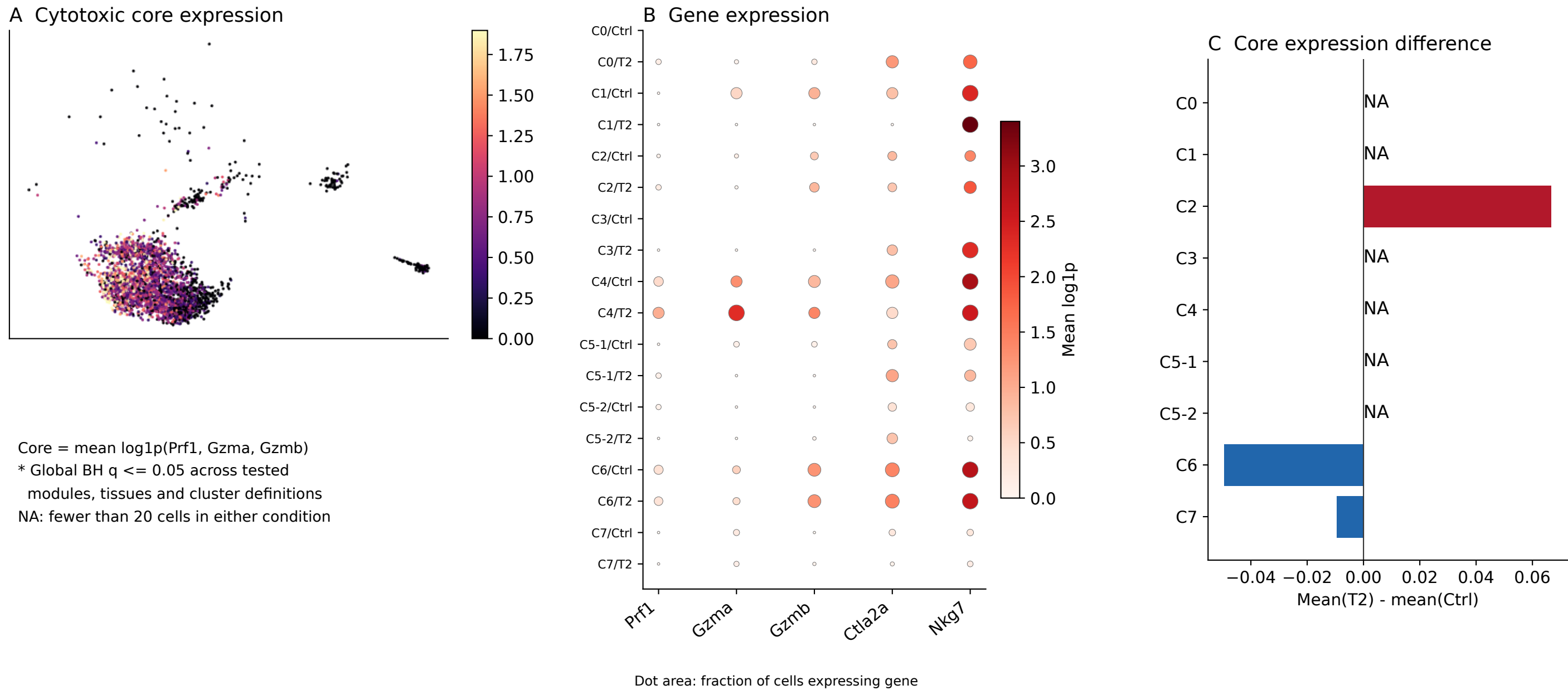
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Thymus: cytotoxic expression and condition effects

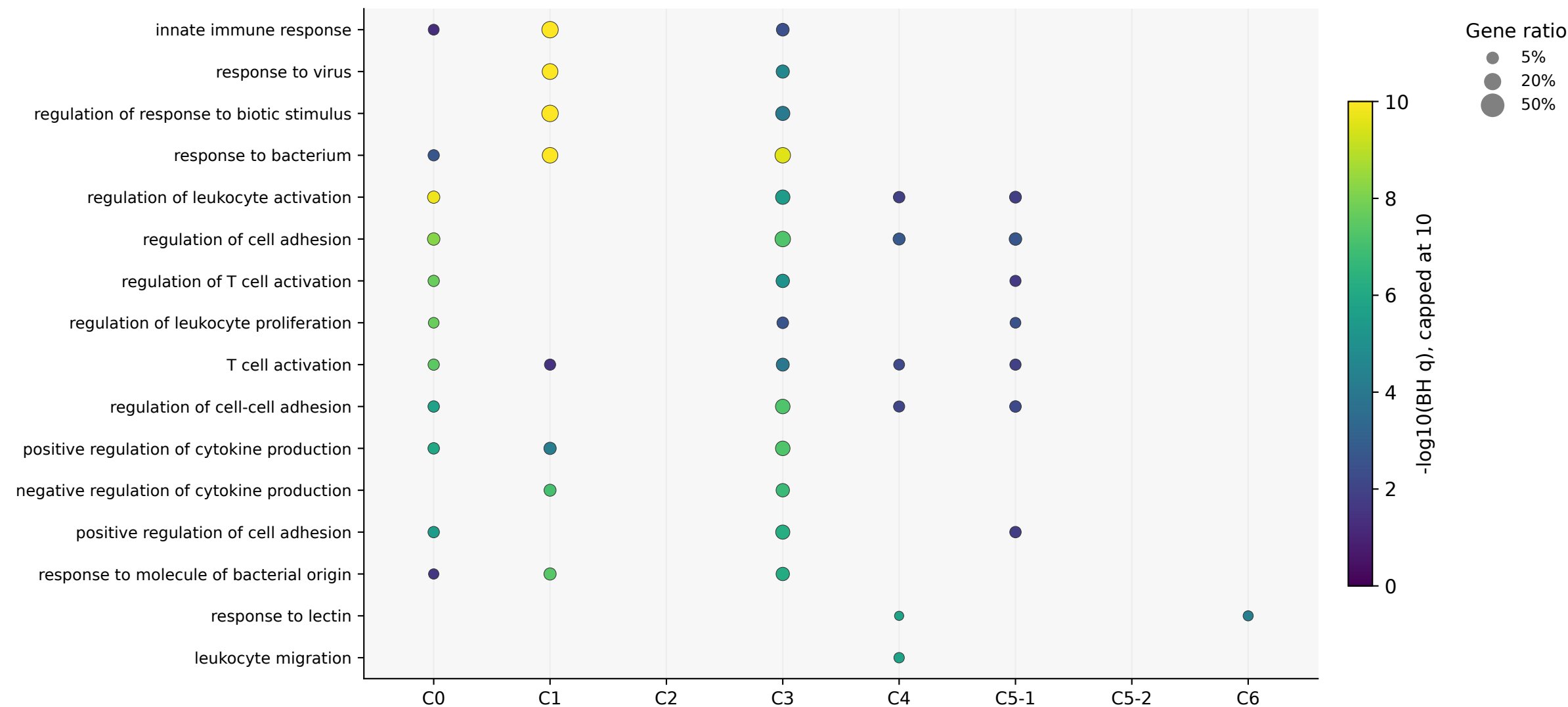
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

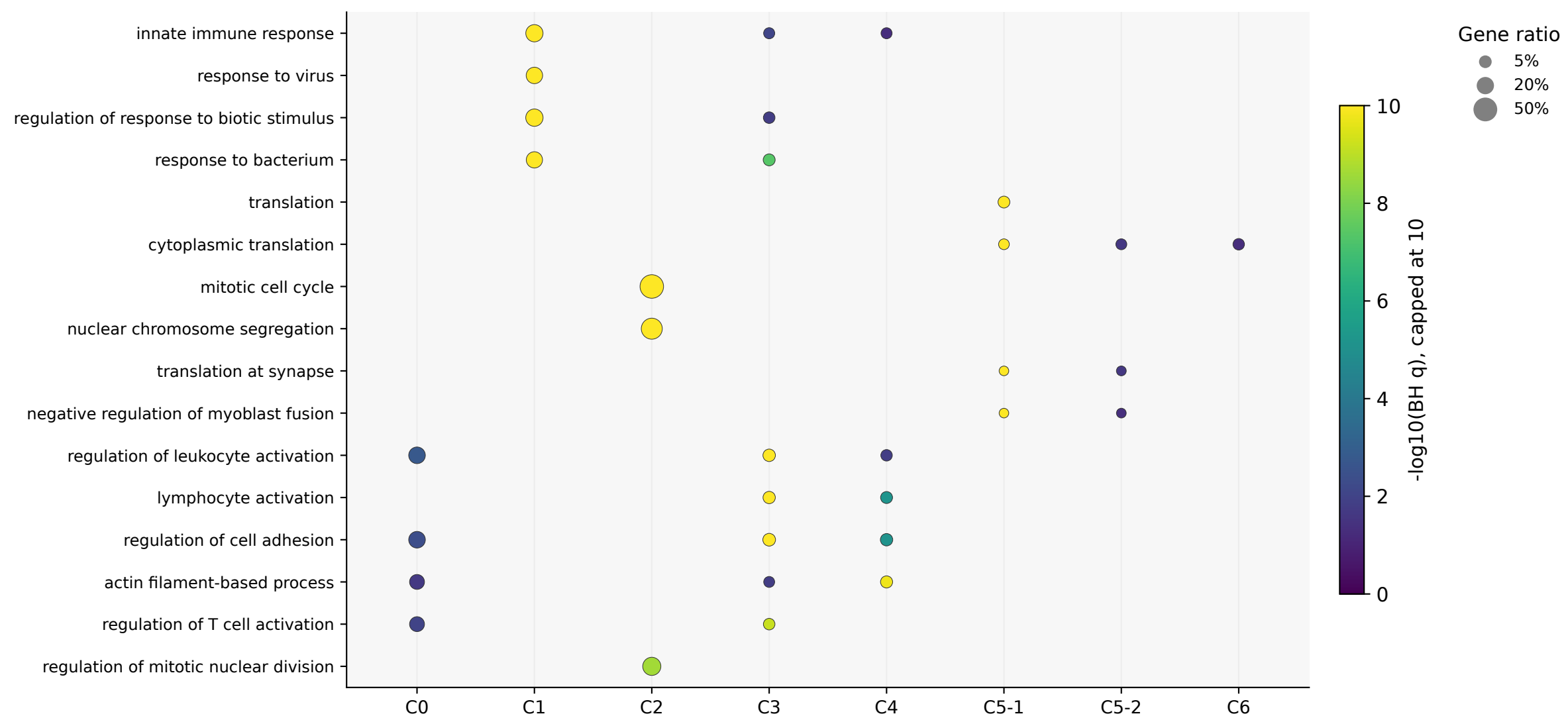


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; q <= .05; >=2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

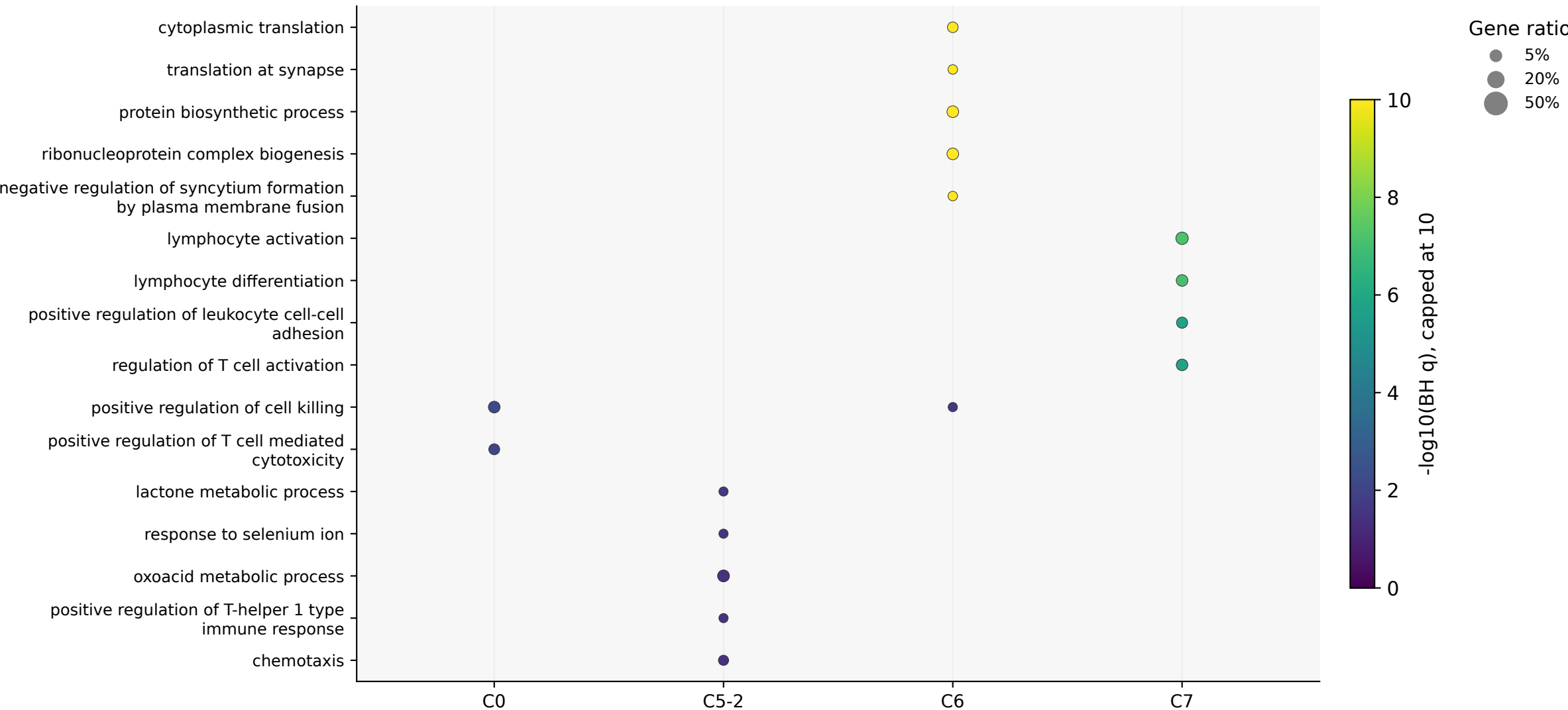


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

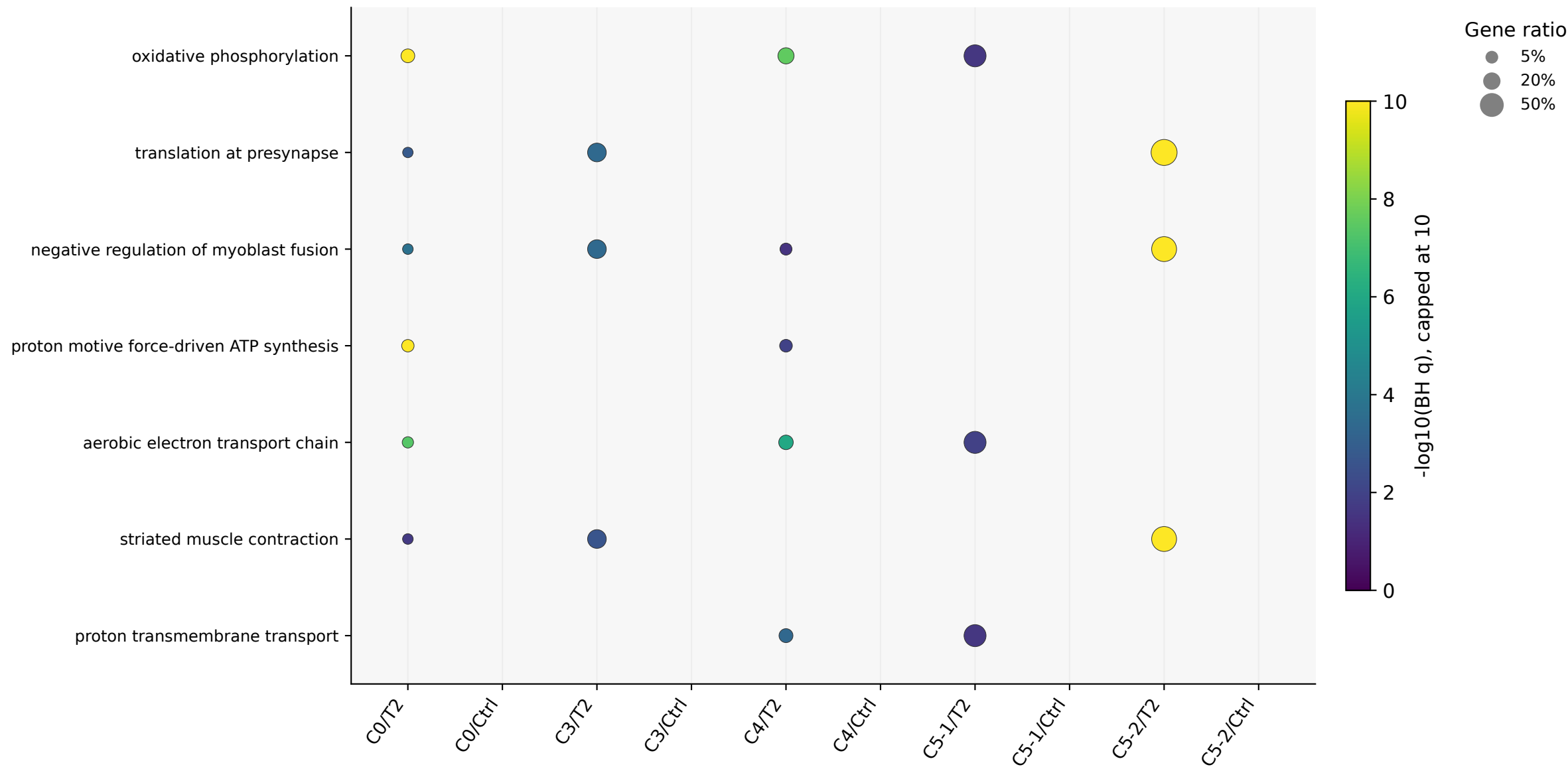


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Bone marrow: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

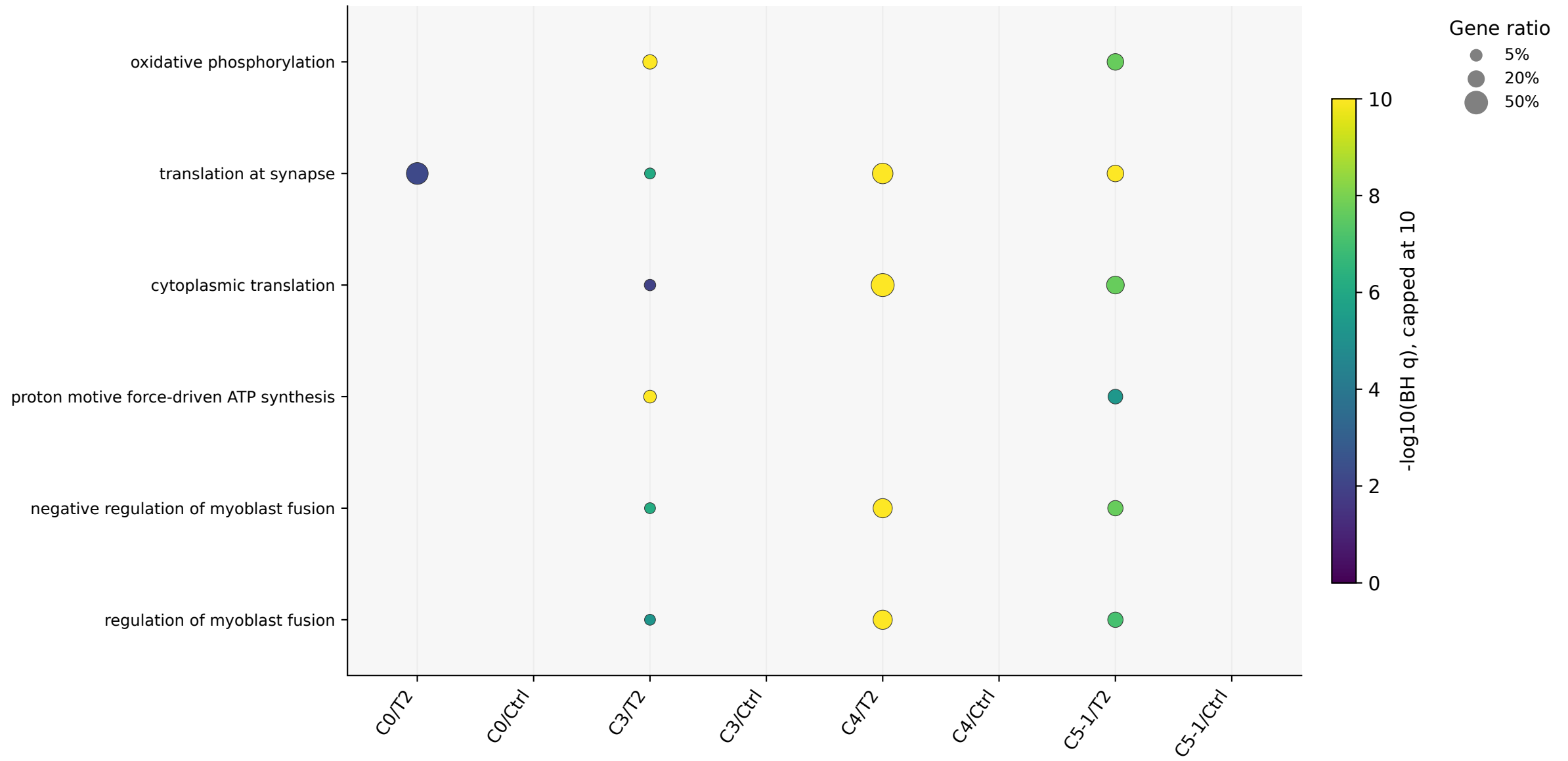


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; q <= .05; >=2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

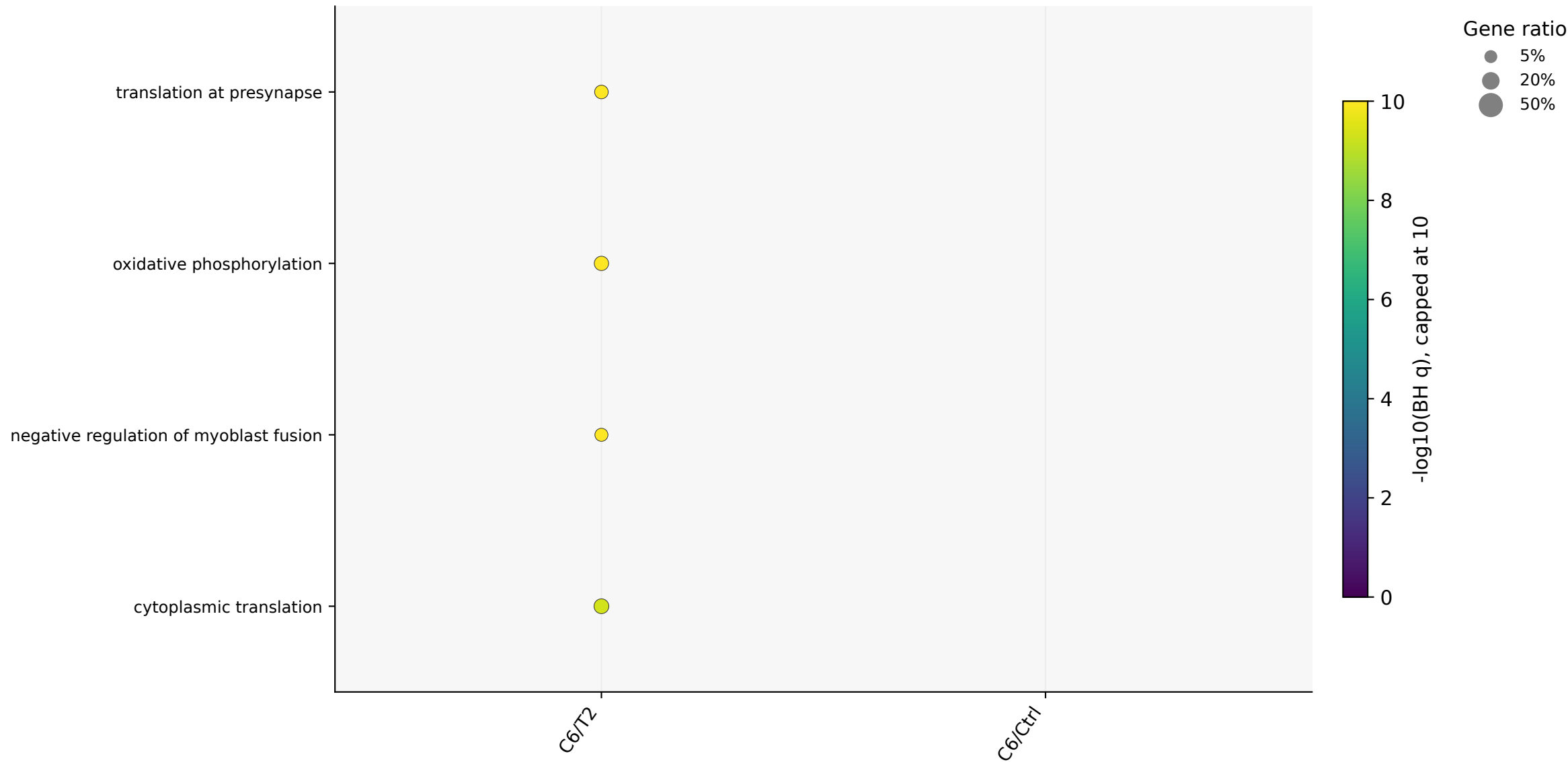


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

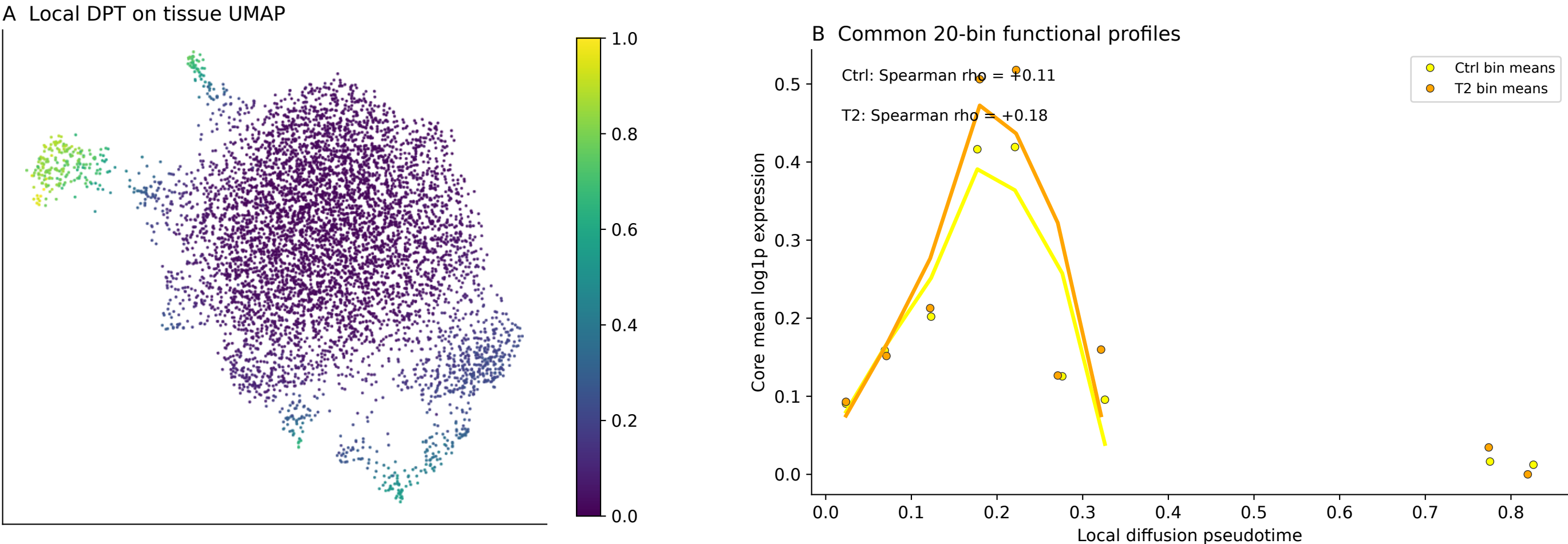


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; q <= .05; >=2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Bone marrow: local cytotoxicity along pseudotime

PAGER-style functional curves on a newly computed within-tissue diffusion map.

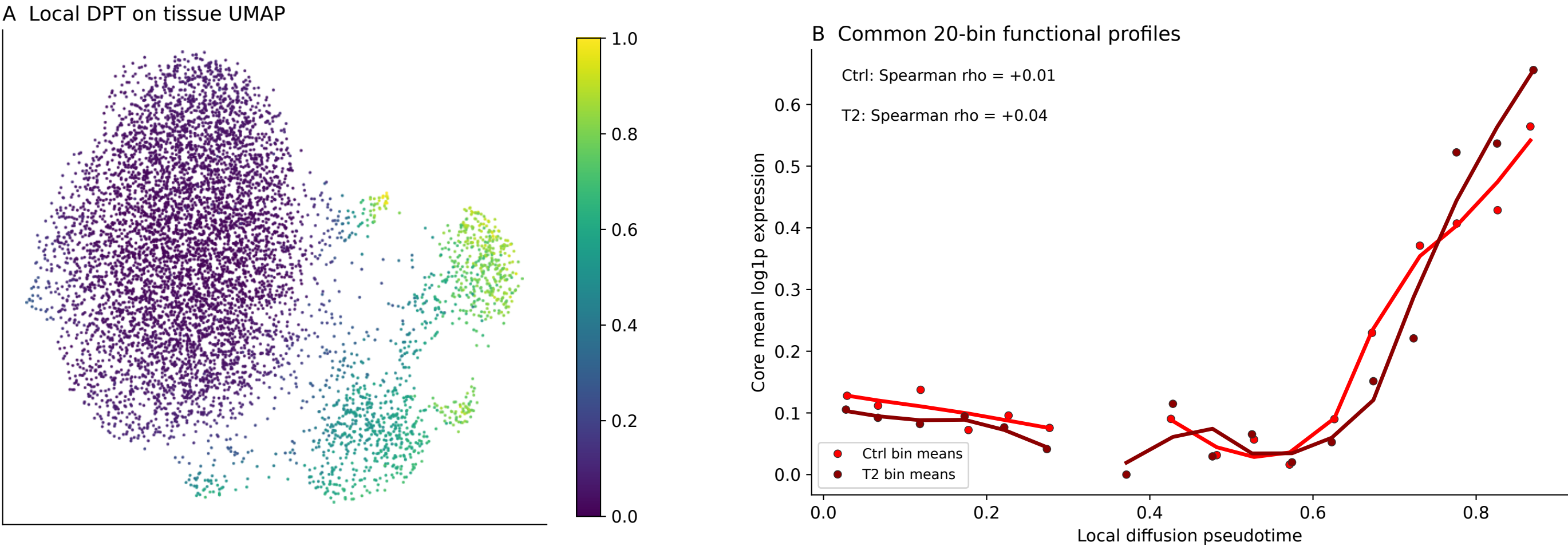


Root state: bone_marrow_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.89, 0.76.
Analyzed 6,110/6,110 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Pseudotime is an assumed ordering, not RNA velocity or lineage validation.

Spleen: local cytotoxicity along pseudotime

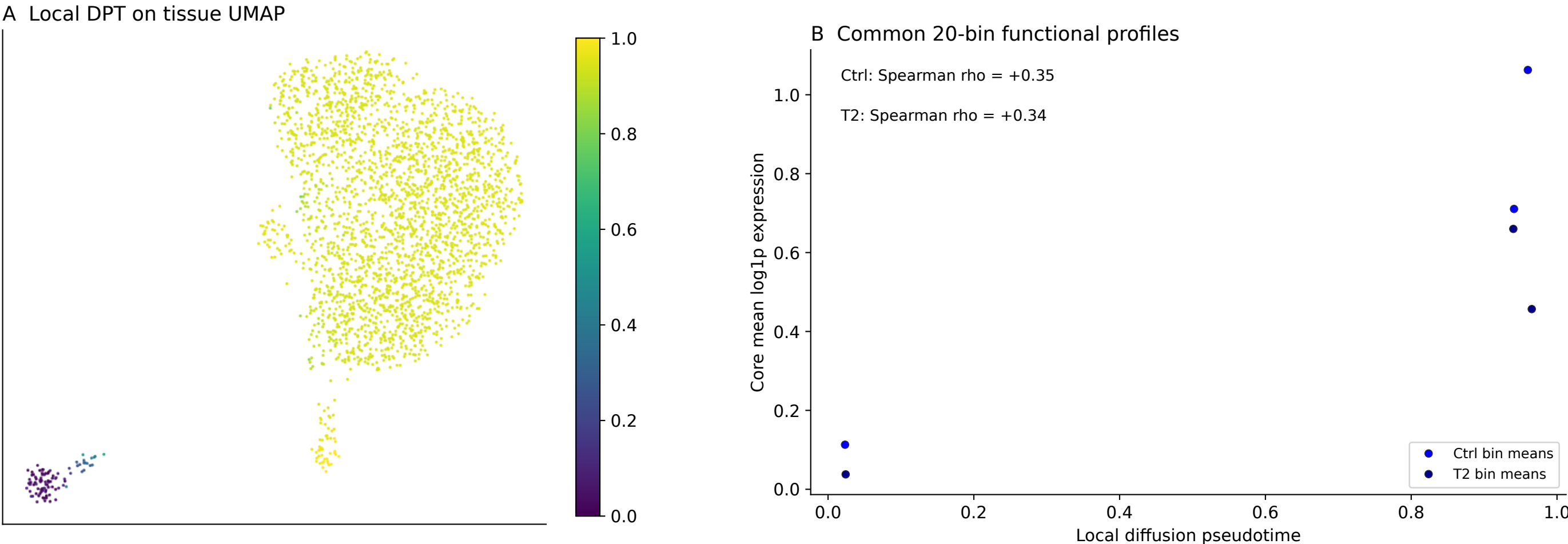
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: spleen_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.97, 0.76.
Analyzed 6,793/6,793 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Thymus: local cytotoxicity along pseudotime

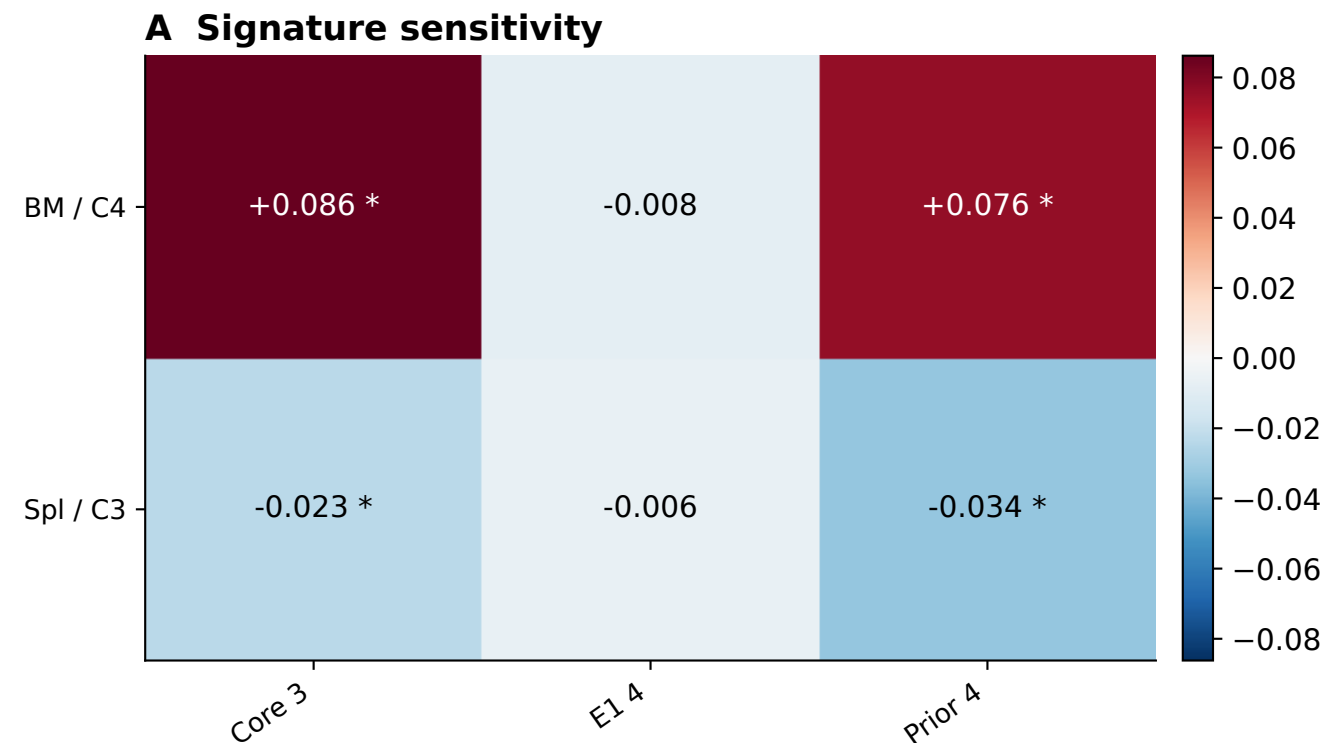
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: thymus_S5; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 1.00, 1.00.
Analyzed 2,629/2,629 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Core signature changes do not imply a full killing pathway

Two reproducible local core signals; the broader E1 signature is not significant.



B Complementary evidence from the same dataset

Bone marrow C4 core: +0.086; q = 0.025
Spleen C3 core: -0.023; q = 0.0015
Both: same direction in all 5 QC resamples

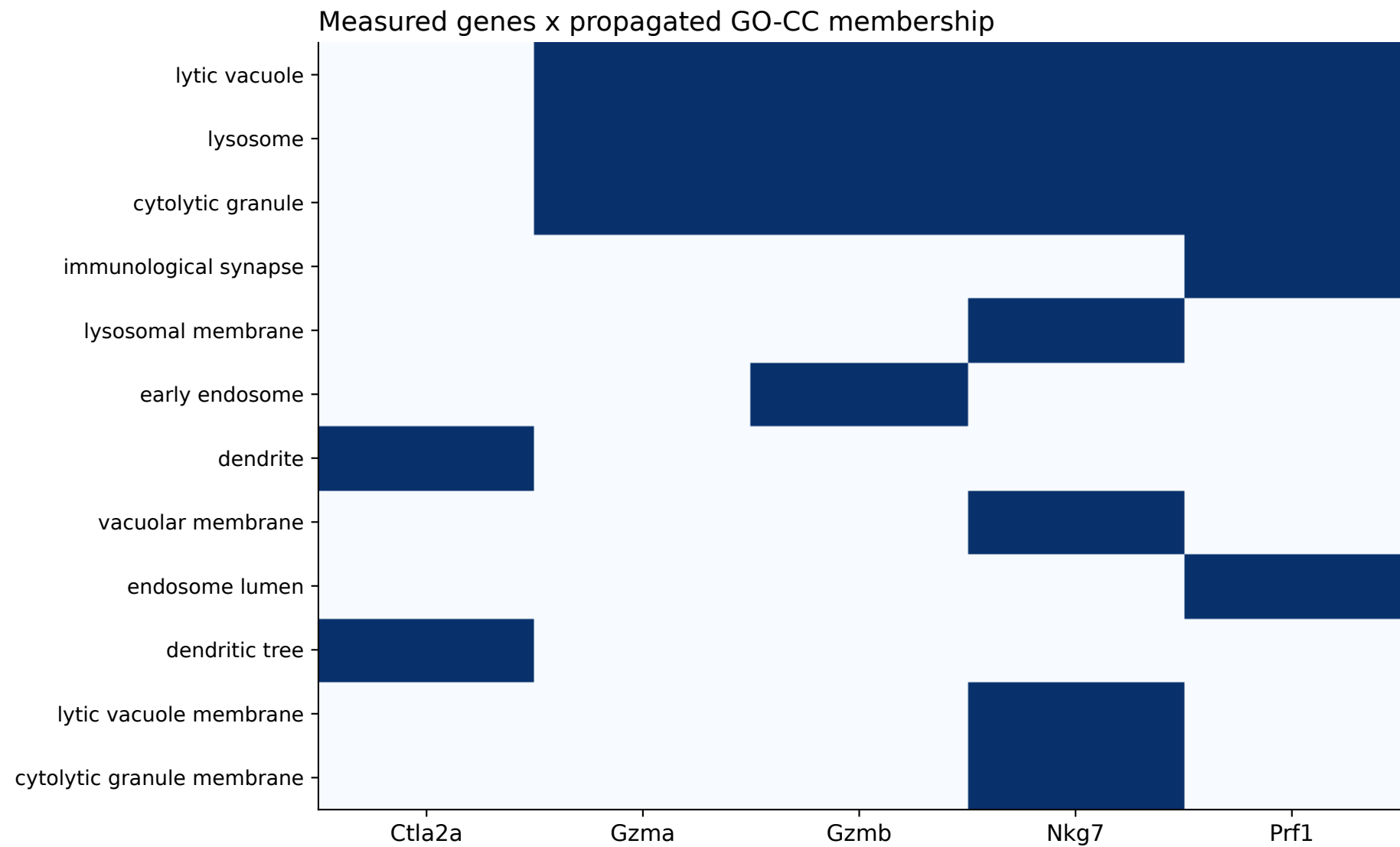
Stable-cluster effects: +0.111 and -0.022
Original-cell overlap: 98.7% and 99.5%

E1 (+Ctla2a): q = 0.93 and 0.53
Killing-related GO terms: no q ≤ 0.05

The evidence supports a localized core-effector transcriptional shift.
It does not establish increased/decreased killing activity or coordinated activation of a full cytotoxic pathway.

Cellular components of cytotoxic module genes

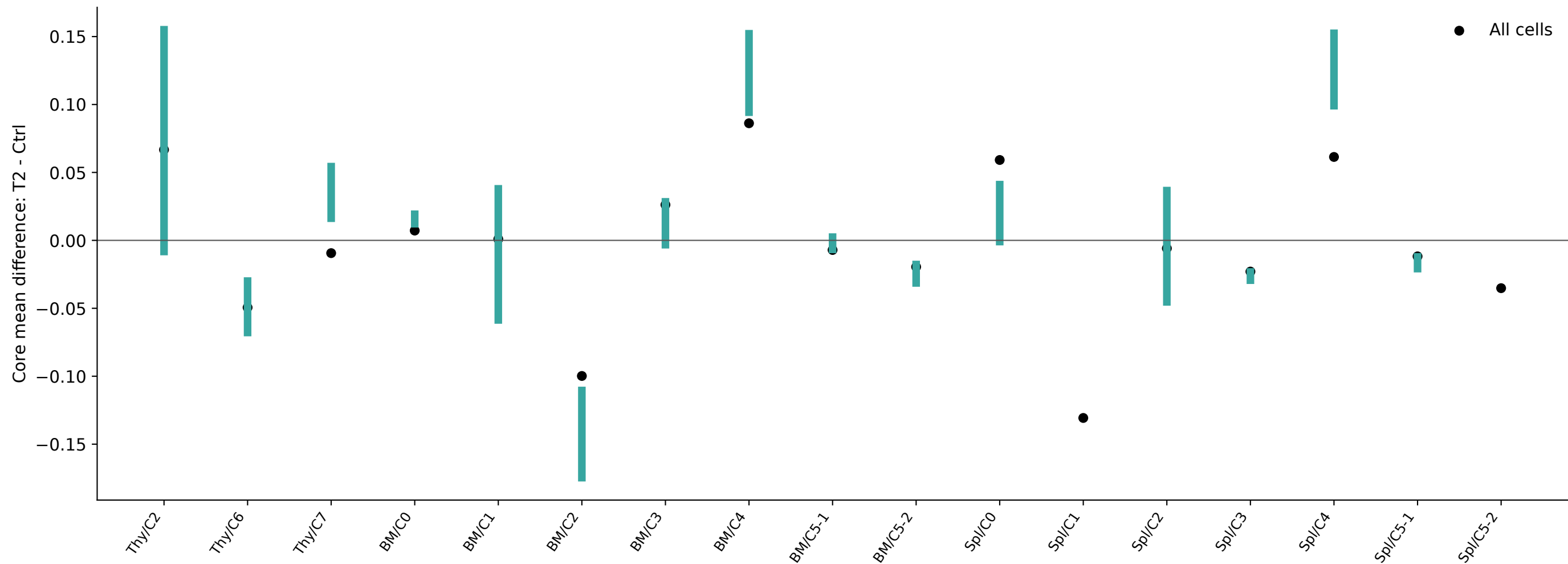
GO-CC membership supports localization; this is not a protein-interaction network.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Robustness: QC matching and alternative clustering

Five condition-balanced QC resamples within each tissue x cluster.



Teal range: five QC-matched effect estimates (not a confidence interval).
Stable tissue clusters are a separate sensitivity analysis; their labels are not interchangeable with original C labels.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Evidence summary and next interpretation

Cytotoxic module results, GO support and sensitivity are reported separately.

Contrast	Core delta	BH q (cell level)	QC same direction	Control-score agrees	Max QC SMD
Spl/C1	-0.131	0.059	NA	yes	NA
BM/C2	-0.100	0.22	5/5	yes	0.17
BM/C4	+0.086	0.025	5/5	yes	0.06
Thy/C2	+0.067	0.49	4/5	yes	0.18
Spl/C4	+0.061	0.3	5/5	yes	0.20
Spl/C0	+0.059	0.46	4/5	yes	0.19
Thy/C6	-0.049	0.4	5/5	yes	0.08
Spl/C5-2	-0.035	0.34	NA	yes	NA

Ranked by absolute expression difference, not selected for significance.
A cytotoxic transcript signature is not a killing assay; GO terms can share the same driver genes.
Independent biological replicates and functional assays are required to confirm treatment effects.